

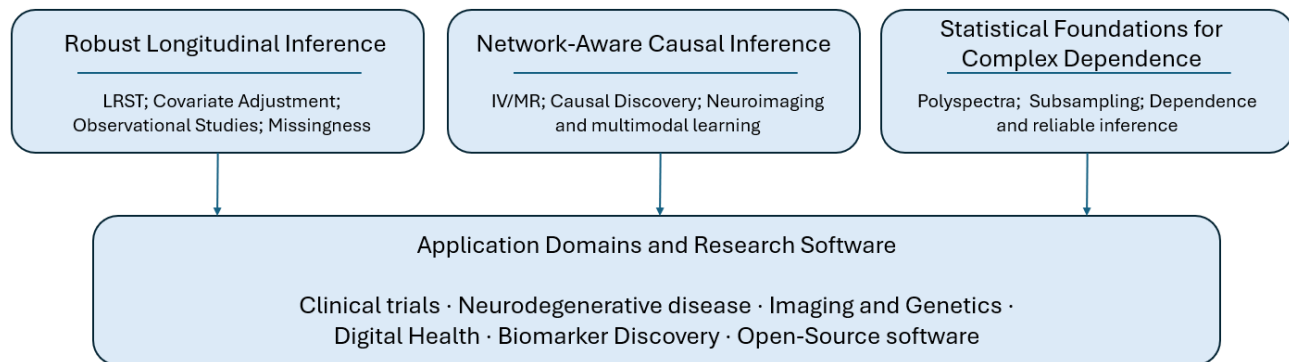
Research Statement

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My research develops statistical methodology for **complex dependent biomedical data**, with particular emphasis on longitudinal studies, causal inference, network-structured data, and neurodegenerative disease. Across these settings, the scientific data are richer than the assumptions built into many classical procedures: outcomes may be multivariate, repeatedly measured, skewed, bounded, or missing; genetic instruments and imaging features are strongly structured; and clinically relevant effects may be distributed across many outcomes rather than concentrated in a single endpoint. My goal is to build methods that remain statistically principled in these settings while being usable in real scientific studies.

Three methodological themes organize my program: **(1) robust inference for multivariate longitudinal and clinical-trial data**, **(2) network-aware causal inference and biomarker discovery**, and **(3) statistical theory for complex dependence**. These themes grew from my doctoral training in time-series and resampling theory, my postdoctoral work in biostatistics at Duke University, and my current independent program at Kennesaw State. They are connected by a common emphasis on nonparametric or semiparametric inference, structured high-dimensional information, and rigorous uncertainty quantification. Clinical trials, neuroimaging, genetics, digital health, and open-source software provide the translational setting in which I develop and test these ideas [1, 2, 3, 4].



1. Robust inference for multivariate longitudinal and clinical-trial data

My first major research program concerns treatment-effect inference when several correlated outcomes are observed repeatedly over time. This setting is common in Alzheimer's disease (AD), Parkinson's disease (PD), and other chronic disorders, where cognition, function, symptoms, and imaging measures evolve jointly. Standard mixed-model and MANOVA-type procedures can perform poorly when outcomes are non-Gaussian, measured on different scales, or affected by outliers and irregular missingness.

With Prof. Sheng Luo at Duke, I developed the **Longitudinal Rank-Sum Test (LRST)**, a nonparametric framework for global treatment-effect testing with multivariate longitudinal outcomes. The two-arm foundation paper, published in the *Statistics in Biopharmaceutical Research* [5], provides a distribution-robust alternative to model-based approaches. A subsequent extension to multi-arm clinical trials has been published in the *Journal of Multivariate Analysis* [6]. I also developed power and sample-size methodology for LRST-based trial design [7]. A related goodness-of-fit test for covariance structures in multivariate longitudinal data is currently under review at *Biometrics* [8]. Together, these projects move LRST from a single testing procedure toward a broader toolkit for complex longitudinal trials.

My current work pushes this program in three directions. First, I am developing **covariate-adjusted LRST procedures** in which nuisance structure is removed through regression or modern machine-learning residualization while preserving valid rank-based inference. Second, I am extending the framework to **observational longitudinal studies**, where treatment or exposure assignment is nonrandom and must be handled through weighting or doubly robust ideas such as IPW, AIPW, and TMLE. Third, I am developing methods for **missing and irregular longitudinal data**, including subject-specific weighting under MAR and sensitivity analyses under departures toward MNAR. These extensions form the core of my next R01-level program: a unified framework for robust longitudinal treatment-effect inference that combines rank methods, machine learning, missing-data robustness, and principled uncertainty quantification [9, 10].

A complementary project develops a functional global test in a reproducing-kernel Hilbert space (RKHS), allowing treatment differences to be detected across a richer class of longitudinal trajectories than those targeted by conventional mean-based tests. This work motivated an NIH R21 application and remains an active methodological direction. More broadly, I am interested in adaptive and sequential versions of these procedures, so that nonparametric global

inference can contribute not only to post-hoc analysis but also to the design of modern multi-endpoint trials. [11].

2. Network-aware causal inference and biomarker discovery

My second research program asks how causal inference should change when exposures, instruments, or outcomes are connected by known biological structure. In imaging genetics, for example, single-nucleotide polymorphisms are correlated through linkage disequilibrium while brain regions are organized by anatomical or functional connectivity. Treating these features as unrelated can discard information and destabilize high-dimensional causal estimation.

With Samhita Pal, I developed **network-aware instrumental-variable regression** methods that incorporate graph structure directly into selection and estimation. The resulting IVGL/IVGL-S framework uses network penalties to borrow information across related variables while allowing invalid instruments to be selected and controlled. The manuscript is currently under major revision at the *Journal of the American Statistical Association* [12]. This methodology is also the statistical core of my revised NIH R15 application, *Network-Aware Mendelian Randomization for Causal Brain Imaging in Alzheimer's and Parkinson's Disease*, on which I am sole PI. The project extends the framework to continuous, binary, multinomial, survival, and repeated-measure outcomes and is designed to identify brain regions that may lie on causal pathways from genetic variation to disease progression.

A closely related line concerns **causal structure discovery**. With Samhita Pal and Shu Yang, I developed a penalized Fast Causal Inference procedure for sparse causal graphs in the presence of latent confounding, with an application to Parkinson's disease biomarker discovery. That work has been accepted at the *Annals of Applied Statistics* and is accompanied by an R package [13]. With student collaborators, I have also explored the interface between causal inference and graph neural networks for AD, resulting in a paper at DATA 2026 [14]. These projects broaden the research question from estimating one causal effect to learning biologically interpretable causal organization among many variables.

I am also extending this program to **distribution-valued and representation-based imaging features**. With Rahul Ghosal, I am developing graph-structured survival models that use regional brain-intensity distributions to predict time to AD [15]; this collaboration underlies an NIH R21 application on which I serve as multiple PI and which is currently pending IRG review. A separate R21 application explored atlas-independent neuroimaging biomarkers derived from foundation models. That project sharpened an important future direction: combining learned representations with explicit statistical structure so that modern imaging features remain interpretable, uncertainty-aware, and portable across cohorts and scanners.

Over the next several years, I plan to connect these strands through **multimodal causal learning**: joint models for genetics, imaging, proteins, and digital phenotypes; network-aware Mendelian randomization with complex outcomes; and causal inference under evolving or latent network structure. I am particularly interested in graphon-based models and network interference, where the network is itself an estimand or source of uncertainty rather than a fixed design object.

3. Statistical foundations for complex dependence

The theoretical foundation of my research comes from work on higher-order dependence, nonlinear time series, and resampling. During my doctoral training with Prof. Soumendra Lahiri and Prof. Tucker McElroy, I developed methodology for **polyspectral means** and higher-order frequency-domain inference for nonlinear processes. This work includes asymptotic theory for polyspectral-mean estimation, accepted in *Bernoulli* [16]; polyspectral factorization in the *Annals of the Institute of Statistical Mathematics* [17]; and quadratic prediction using auto-cumulants in *Sankhyā A* [18]. I have also developed software for polyspectral estimation and quadratic forecasting.

A current project studies subsampling and convolved-bootstrap inference for polyspectral means [19]. More broadly, this work gives me a theoretical perspective that carries directly into my biomedical methodology: I am interested in procedures that remain valid under dependence, non-Gaussianity, high dimensionality, and imperfect model specification. Rather than treating theory as a separate line from applied biostatistics, I use it to identify where standard approximations fail and to design inferential procedures with transparent operating characteristics. [20, 21].

Funding strategy, collaboration, and research training

External funding is an integral part of my research strategy. Since beginning my faculty appointment, I have submitted **five NIH applications as PI or multiple PI**, including a revised R15 and an MPI R21 that are currently pending review. Three additional NIH applications have completed peer review, and I have also participated as co-investigator on three interdisciplinary proposals led by collaborators outside statistics. I treat grant development as an iterative part of the research process: reviewer feedback helps refine the scientific question, strengthen preliminary evidence, and resubmit the strongest projects. My immediate goal is to convert this portfolio into sustained R15/R21 support and then into an R01-level program centered on robust longitudinal inference and network-aware causal methods.

This program is collaborative by design. My current network includes collaborators in biostatistics, radiology,

neurology, computer science, and public health at Duke, South Carolina, Texas A&M, NC State, and Kennesaw State. These collaborations provide clinically meaningful questions and data, while also creating settings in which methodological ideas can be stress-tested beyond the datasets that originally motivated them.

Research training is equally central to the program. In my first year as faculty, I mentored undergraduate and master's researchers whose projects progressed to peer-reviewed outputs. Three undergraduates supported through Kennesaw State's First-Year Scholars Program produced two abstracts accepted at NCUR, a short paper accepted at ICMLA [22], a journal manuscript receiving a favorable revision at the *American Journal of Undergraduate Research* [23], and another manuscript now being prepared for submission. Two of these students continued into the Sophomore Scholars Program. I also mentored master's students on methodological and biomedical projects; one student co-authored a DATA 2026 paper on causal inference and graph neural networks for AD [14], another manuscript is nearing submission, and one master's student has continued with me into the PhD program. I currently advise doctoral students as well. My goal is to build a research pipeline in which students progress from reproducible analysis to methodological ownership and, ultimately, independent scholarship.

Five-year research agenda

My next five years will focus on three connected objectives.

Robust longitudinal inference beyond randomized trials. I will develop a general framework that links rank-based global testing with causal adjustment for observational studies. Methodologically, this requires understanding when data-adaptive residualization or nuisance-function estimation can be combined with ranks without compromising type-I error, how IPW/AIPW/TMLE ideas can be incorporated while retaining robustness to outcome-model misspecification, and how subject-specific weighting should be constructed under intermittent and informative missingness. I plan to study these questions theoretically and through simulation regimes that deliberately separate smooth, nonlinear, discontinuous, and high-dimensional nuisance structures. The goal is not simply another LRST variant, but a reusable inferential framework for comparative effectiveness studies and pragmatic trials with multivariate longitudinal outcomes. This program is the foundation of the R01 application I am now developing.

Multimodal and network-aware causal learning. I will extend network-aware IV and causal-discovery methods to settings in which genetics, imaging, proteomics, and digital phenotypes are observed jointly. Important methodological problems include invalid instruments, high-dimensional weak signals, latent confounding, and disagreement between anatomical networks and data-learned representations. I am especially interested in combining interpretable statistical structure with modern representation learning: foundation-model or graph-neural-network features can be useful, but the inferential target, uncertainty, and biological interpretation must remain explicit. A parallel theoretical direction concerns causal inference under network interference and graphon-type latent structure, where the observed network is noisy, evolving, or only indirectly measured.

Theory and computation for reliable modern inference. As these methods become more adaptive, theoretical guarantees become more important rather than less. I will continue developing resampling and dependence-aware asymptotics for statistics involving ranks, learned nuisance functions, nonlinear dependence, and structured high-dimensional estimators. I am also interested in computational tools that make complex methodology reproducible: simulation engines for method validation, open-source implementations, and eventually AI-assisted workflows for curating trial protocols and generating transparent simulation designs. Here AI is not the inferential endpoint, but a tool for making statistical development more systematic, auditable, and scalable.

Open-source software will remain a first-class research output. I currently maintain or co-develop software for penalized causal discovery, network-aware IV regression, polyspectral analysis, and quadratic forecasting. Over the next several years I plan to consolidate method-specific code into documented CRAN/Bioconductor-quality tools with reproducible examples drawn from clinical-trial and neuroimaging applications. This software strategy also creates natural training projects through which students can learn both statistical methodology and research software engineering.

The long-term goal is a research program in which statistical theory, methodological innovation, biomedical translation, and student training reinforce one another. I want to develop methods that are rigorous enough to advance statistical science, flexible enough to address modern biomedical data, and accessible enough to be used by collaborators. At a research-intensive Statistics or Biostatistics department, I would contribute a program spanning longitudinal methods, causal inference, high-dimensional and network data, and computational statistics, while building collaborations in neuroscience, clinical research, public health, and data science. That combination, i.e. robust inference, causal structure, complex dependence, and translational collaboration, defines the research program I plan to build over the next stage of my career.

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